

Integrating Spatial Design with Reality: An AR Game Scene Generation Approach

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ABSTRACT

AR games can offer a distinct and immersive experience set apart from traditional games. When creating AR games, one of the most formidable challenges faced by designers lies in the unpredictability of real-world environments, making it difficult to seamlessly integrate virtual world with the player's surroundings. Our approach addresses this challenge by converting the designer's design and the player's real environment into scene graphs, and then integrating the two graphs to determine the placement of virtual objects. In addition, we provide a user-friendly interface for designers to quickly visualize inspirations, and an integration module for naturally arranging virtual objects.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

Over recent years, Augmented Reality (AR) has become a prominent field with diverse applications ranging from retail [1] to education [2]. One of its most captivating applications lies in the domain of games, where the integration of virtual objects into real world environments opens new avenues for interactive and immersive experiences. Different from traditional video games, which are confined to preset virtual environments, AR games actively incorporate real world contexts such as player movements, surroundings, and activities of other players. Widely known games such as *Pokemon Go* [5], *Ingress Prime* [6], and *Minecraft Earth* [4] serve as very good evidence of the rapidly increasing popularity of AR in gaming.

In order to provide immersive gaming experiences, AR games should naturally integrate the real environment with game elements and effectively turn surroundings into an interactive game scene. However, creating immersive AR games poses challenges in naturally integrating virtual and real elements. It's nearly impossible for designers to create tailor-made AR scenes for all the players, especially when the players' environments are unknown. A comprehensive approach is not only needed for AR games to fully leverage the relationship between virtual content and the real world, but also essential to assist designers in flexibly realizing their concepts across diverse scenarios. To fulfill these considerations, we proposed a graph-based approach that includes a designer-friendly interface enabling intuitive visualization, and an integration module enabling arrangement of game elements.

Previous works utilizing machine learning show great promise in enabling good integration. For example, Tahara et al. [8] con-

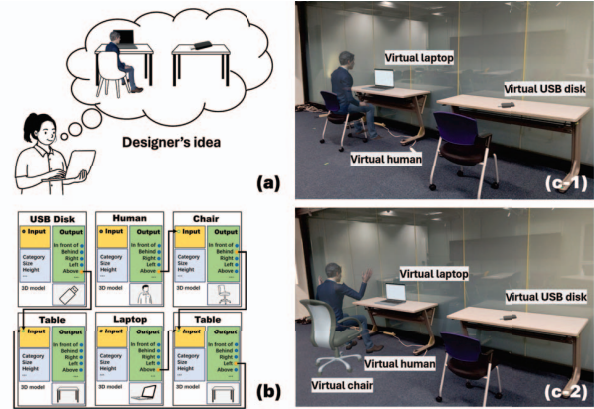


Figure 1: Illustration shows how our system realizes a designer's idea in different real environments, including a case in (c-2) where real chair is missing, our approach can generate a virtual one.

verted the classification results obtained from neural networks and relationships of real objects into scene graphs, match with predefined AR contexts to generate AR scenes. Liang et al. [3] employed deep learning to capture behavior sequences of real cats, and also used neural networks for semantic segmentation to associate these behaviors with real-world objects. Qian et al. [7] built a database from a vast collection of designs created by designers in Virtual Reality, facilitating a data-driven approach to bind virtual objects with real-world environments.

Our work explored a new way to enhance the AR game scene creation, as shown in Fig. 1. We employed a designer-centric approach to seamlessly integrate imaginative blueprints with real-world environments. Specifically, we developed a graph-based interface for designers, enabling them to effortlessly translate their creative ideas into detailed game scene templates. Furthermore, our approach developed an algorithm to match the scene graph from templates with the scene graph from actual environments. In addition, we also developed a virtual object arrangement method for realizing the proposed AR scene that supports various placement strategies. Our technique provides a flexible and intuitive method to create AR game scenes.

2 METHODS

This section will present our proposed approach for creating immersive AR game scenes. An overview of our proposed approach is shown in Fig. 2.

2.1 Game Scene Graph Creation

We introduced a graph-based development interface for game scene creation, which we call Blueprint Interface, shown in Fig. 3. It uses a graph with nodes and edges, where nodes represent objects and their connections define spatial relationships. This interface also

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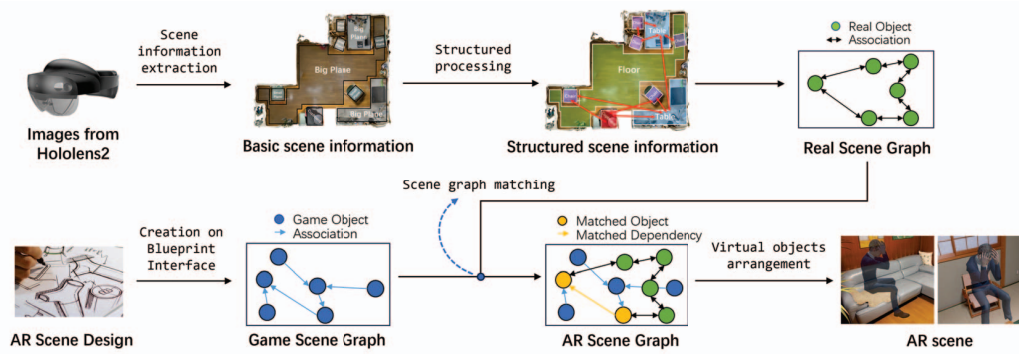


Figure 2: An overview of our approach. There are 2 inputs of our approach, the first input is the RGB-D images captured by HoloLens 2's camera, and the second input is the design of the designers. Then through scene graph matching, we flexibly combine these 2 inputs to reach an immersive AR game scene.

includes object attributes like size, height, and model prefabs, ensuring detailed control over each element in the scene. Relationships between objects are also customizable, with options for specifying exact or fuzzy positional relations.

2.2 Real Scene Graph Creation

Our approach first understands real scene semantics by extracting semantic labels and oriented bounding boxes from point clouds. This involves analyzing planes in the point cloud data, assigning semantic labels, and determining object categories based on orientation, size, and height. To understand spatial relationships, We use a geometry-based method to judge the relationship between objects by their positions. This process results in the creation of the real scene graph. In this graph, nodes represent each object identified and labeled in the real scene, and edges symbolize the spatial relationships between these objects.

2.3 AR Scene Graph Proposal

This involves a process of matching triplets from the game scene graph with those in the real scene graph, taking into account spatial relationships and object categories relevance. Additionally, We validate these matches, ensuring consistency and resolving conflicts by selectively discarding less suitable triplets. This process helps us achieve an optimal mapping of virtual objects in the real world, forming a cohesive and immersive AR game scene.

2.4 AR Content Arrangement in Real Scene

In this stage, our approach generated the AR scene by placing virtual objects based on their relationships with real objects and previously positioned virtual objects. It considers object placements to avoid collisions and maintain spatial harmony. The approach offers several strategies to resolve space constraints, including finding new scene proposals or adjusting spatial relationships and sizes, thus ensuring a seamless and realistic integration of AR content into the real scene.

3 CONCLUSION AND FUTURE WORK

In this work, we introduced a novel graph-based interface along with an integration technique, aimed at simplifying the process for designers to develop AR designs capable of adapting to diverse environments. Future work includes integrating data-driven models for enhanced scene understanding. This research contributes significantly to AR game development, simplifying design processes for creators and enhancing the immersive experience for players.

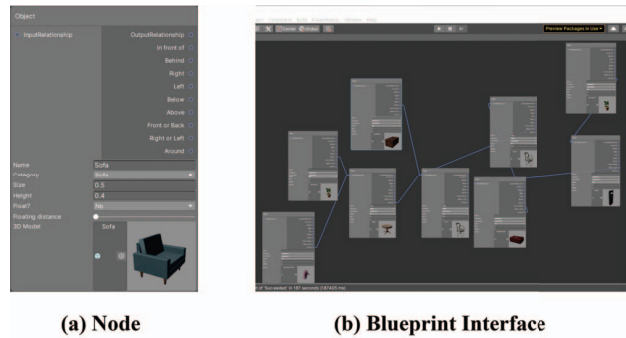


Figure 3: Screenshot of our proposed Blueprint Interface. (a) A screenshot of a node in the interface. (b) The whole interface consists of many nodes.

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